

Liyun Zhang

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EDUCATION

Northeastern University, Boston, MA

Expected May 2027

Master of Electrical and Computer Engineering

GPA: 4.0

Relevant Coursework: Mobile Robotics, Machine Learning for Embedded Systems

Xidian University, Xi'an, China

Jul. 2023

Bachelor of Electronic Information Engineering

TECHNICAL SKILLS

Programming:	C/C++, Python, Lua, Perl, ARM Assembly, LabVIEW
Embedded Systems:	Embedded Linux, Zephyr RTOS, nRF Connect SDK; I2C, SPI, UART, PWM
Development & Debugging:	Git, CMake, Make, Wireshark, tcpdump; TCP/IP, Wi-Fi, Bluetooth
Robotics & ML:	DORA, OpenCV, ORB-SLAM3, VGGT-SLAM, Kalibr, Quest VR Teleoperation, VLA Fine-Tuning

WORK EXPERIENCE

TP-Link Technologies Company, Shenzhen, China

Jul. 2023 – Sep. 2024

Embedded Software Engineer

- Developed **C software for embedded Linux** across kernel and user space. Implemented product features and adapted code to multiple hardware platforms.
- Investigated failures reported during validation and customer use through system logs and network captures. Identified root causes and fixed connectivity and platform-specific software issues.
- Built a crash-diagnostics framework that preserved kernel and application logs through crashes and reboots. Engineers could inspect failure data even when an issue could not be reproduced locally.
- Implemented a multithreaded logging pipeline with a **lock-free ring buffer** and shared logging library. Defined diagnostic data requirements with firmware teams. Deployment in a commercial product cut validation debugging time by at least **30%**.

RESEARCH & PROJECTS

Visual SLAM for Robot Demonstration Collection

Aug. 2026 – Present

The Helping Hands Lab, Northeastern University

- Built a SLAM benchmark using an **Intel RealSense D455** for close-range robot manipulation, covering occlusions and low-texture scenes.
- Compared **ORB-SLAM3** and **VGGT-SLAM** under the same evaluation protocol. ORB-SLAM3 held 5.3 mm translation ATE on drawer-manipulation trajectories, while stereo VGGT-SLAM was roughly $2\times$ worse
- Investigating how known camera intrinsics and fixed stereo geometry can constrain VGGT-SLAM pose and scale estimates. Planning to integrate the method into an **UMI-like** data collection system for more accurate demonstration trajectories.

Retriever: Bimanual Robot Deployment on PiPER-X

Mar. 2026 – Aug. 2026

The Helping Hands Lab, Northeastern University

Project Website

- Built and calibrated a dual-arm **PiPER-X** platform with **Quest VR** teleoperation and **DORA**. Helped collect and process **200–300 demonstration episodes** for drawer-search and bag-retrieval tasks.
- Improved task success rates through repeated **pi-0.5** ($\pi_{0.5}$) fine-tuning and real-robot testing. Used failed executions to revise demonstration collection and retrained policies on the updated data.
- Tuned action smoothing to reduce abrupt reversals and backtracking during deployment. Robot motion became smoother and faster.
- Integrated **Gemini** planning with language-conditioned skills, scene memory, and execution monitoring. The planner selected skill prompts from observations and task progress while robot control continued asynchronously.
- Validated drawer-search and seasoning sequences from object search through retrieval, dispensing, and return. The full system achieved **73% task progress** on spice search and **100%** on bag retrieval.

ActQuant: VLA Quantization on UR3

May 2026 – Jul. 2026

ActQuant: Sub-4-bit Action-Guided Quantization for Vision-Language-Action Models

Manuscript under review at NeurIPS 2026

- Collected manipulation demonstrations and evaluated baseline and ActQuant-quantized vision-language-action (VLA) policies on a **UR3** across **4 tasks**, with **10 trials per task for each method**. The quantized vision and language backbone averaged **3.0 bits per weight**.
- Evaluated **pi-0.5: 75.0% success vs. 77.5% for FP16**, with memory cut from **6.7 GB to 2.7 GB**.